

Laboratory Operations

Complete Visual MCQ Reviewer

Calculations • Quality Control • Statistics • Tables • Graphs • Graphics

Part I - Core Questions

1. A 4% w/v solution is needed in a final volume of 300 mL. How much solute is required?

- A. 4 g
- B. 8 g
- C. 12 g
- D. 16 g

2. A laboratory prepares 750 mL of a 16% v/v solution. What volume of liquid solute is required?

- A. 12 mL
- B. 60 mL
- C. 120 mL
- D. 160 mL

3. Which equation is used for dilution of a concentrated stock?

- A. $C_1V_1 = C_2V_2$
- B. $CV = \text{mean}/SD$
- C. $F = \text{mean}/\text{variance}$
- D. $pH = pK_a - \log \text{salt/acid}$

4. A strong acid has $[H^+] = 2.0 \times 10^{-4} \text{ M}$. Its pH is closest to:

- A. 2.30
- B. 3.70
- C. 4.70
- D. 10.30

5. A solution has pOH 5.2 at 25 C. What is its pH?

- A. 5.2
- B. 7.0
- C. 8.8
- D. 14.0

6. Which equation is appropriate for a weak-acid/conjugate-base buffer?

- A. Henderson-Hasselbalch
- B. F test
- C. Westgard equation
- D. Beer ratio only

7. A patient absorbance is 0.72 and a 240 mg/dL standard reads 0.96. What is the patient concentration?

- A. 120
- B. 180
- C. 240
- D. 320

8. What should be done with a measured serum-blank absorbance before a patient concentration is calculated?

- A. Subtract it from patient absorbance
- B. Add it
- C. Ignore the standard
- D. Double it

9. One international unit of enzyme activity corresponds to:

- A. 1 micromole substrate consumed or product formed per minute
- B. 1 mole per second
- C. 1 mg/L enzyme
- D. 1 absorbance unit/hour

10. Which statistic best compares relative precision at different analyte concentrations?

- A. CV
- B. Bias
- C. Sensitivity
- D. Intercept

11. Approximately what percentage of Gaussian QC values fall within ± 2 SD?

- A. 68%
- B. 90%
- C. 95%
- D. 99.7%

12. A single control exceeds ± 2 SD. Which warning rule is triggered?

- A. 1_{2s}
- B. 2_{2s}
- C. R_{4s}
- D. 4_{1s}

13. Two consecutive controls exceed the same +2 SD boundary. Which rule is violated?

- A. 1_3s
- B. 2_2s
- C. R_4s
- D. 10x

14. One control exceeds +3 SD. Which rule applies?

- A. 1_3s
- B. R_4s
- C. 4_1s
- D. F

15. Four consecutive controls exceed the same 1 SD boundary. Which rule is implicated?

- A. 4_1s
- B. R_4s
- C. 1_3s
- D. 1_2s

16. A progressive rise over several QC runs is called a:

- A. Trend
- B. Shift
- C. Random spike
- D. Reference interval

17. Several consecutive QC values stay on the same side of the mean after a system change. This is a:

- A. Shift
- B. Random error
- C. Normal scatter
- D. R4s event

18. Which plot is the standard sequential display of QC values around mean and SD limits?

- A. Levy-Jennings
- B. ROC
- C. Bias plot
- D. Regression

19. Which plot is used for interlaboratory comparison of two control levels?

- A. Tonks-Youden
- B. ROC
- C. Histogram
- D. Bias

20. Which method accumulates small QC deviations to detect early systematic movement?

- A. CUSUM
- B. F test
- C. Reference interval
- D. Sensitivity

21. Which statistical test compares two means?

- A. Student t
- B. F
- C. ROC
- D. CV

22. Which test compares variances?

- A. F test
- B. t test
- C. SDI
- D. Sensitivity

23. A candidate mean is 206 and reference mean 200. Bias equals:

- A. +6
- B. -6
- C. +3
- D. 0

24. Error that increases with concentration is:

- A. Proportional
- B. Constant
- C. Random only
- D. Delta

25. A regression intercept different from zero estimates:

- A. Constant error
- B. Proportional error
- C. Prevalence
- D. Precision

26. A regression slope different from 1 estimates:

- A. Proportional error
- B. Constant error
- C. Sensitivity
- D. Random error

27. Specificity is:

- A. $TN/(TN+FP)$
- B. $TP/(TP+FN)$
- C. $TP/(TP+FP)$
- D. $TN/(TN+FN)$

28. Sensitivity is:

- A. $TP/(TP+FN)$
- B. $TN/(TN+FP)$
- C. $TP/(TP+FP)$
- D. $FP/(TN+FP)$

29. When establishing a nonparametric reference interval directly, the source recommends at least:

- A. 20 people
- B. 40
- C. 60
- D. 120

30. Which statistic compares a lab mean with its peer mean in SD units?

- A. SDI
- B. CV
- C. F
- D. PVP

31. Which tool can reveal possible specimen misidentification from an unexpectedly large change in serial patient results?

- A. Delta check
- B. CUSUM
- C. ROC
- D. Youden

32. Which framework expresses analytical defects per million opportunities?

- A. Six Sigma
- B. Beer law
- C. Henderson-Hasselbalch
- D. t test

33. A new reagent lot is introduced. Which action best matches the chapter?

- A. Compare new and current lots using controls and patient samples
- B. No QC needed
- C. Use one patient only
- D. Change reference interval automatically

34. An IQCP risk assessment includes error likelihood and:

- A. Severity of harm
- B. Reagent cost
- C. Number of shifts
- D. Prevalence only

35. $TEa=12$, $bias=2$, $SD=2$. Sigma equals:

- A. 2
- B. 4
- C. 5
- D. 7

Part II - Table-Based Questions

36. Use the table. Which assay has the poorest relative precision?

Analyte	Mean	SD
A	2.5	0.30
B	4.0	0.20
C	100	2.0
D	140	2.8

- A. A
- B. B
- C. C
- D. D

37. Use the method table. Which method shows positive bias?

Method	Mean	SD
Reference	235	3.8
Candidate	246	3.4

- A. Reference
- B. Candidate
- C. Neither
- D. Cannot determine

38. Use the diagnostic table. What is sensitivity?

	Disease +	Disease -
Test +	42	8
Test -	6	44

- A. 84%
- B. 87.5%
- C. 88%
- D. 92%

39. Using the same table, what is specificity?

	Disease +	Disease -
Test +	42	8
Test -	6	44

- A. 80%
- B. 84.6%

- C. 88%
- D. 91.7%

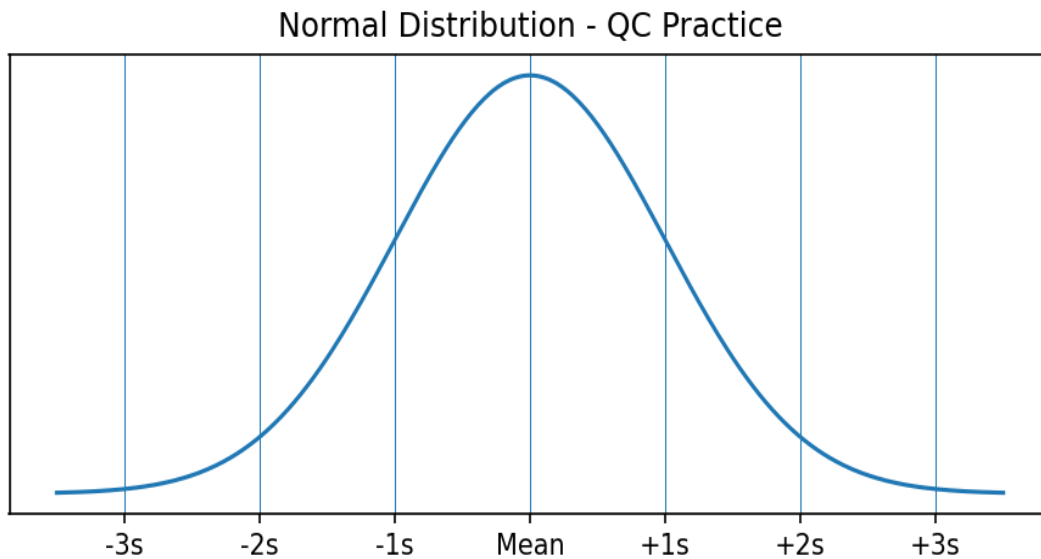
40. A laboratory compares two osmometers. Which has the lower CV?

Instrument	Mean	SD
A	280	3.1
B	294	2.8

- A. A
- B. B
- C. Equal
- D. Cannot determine

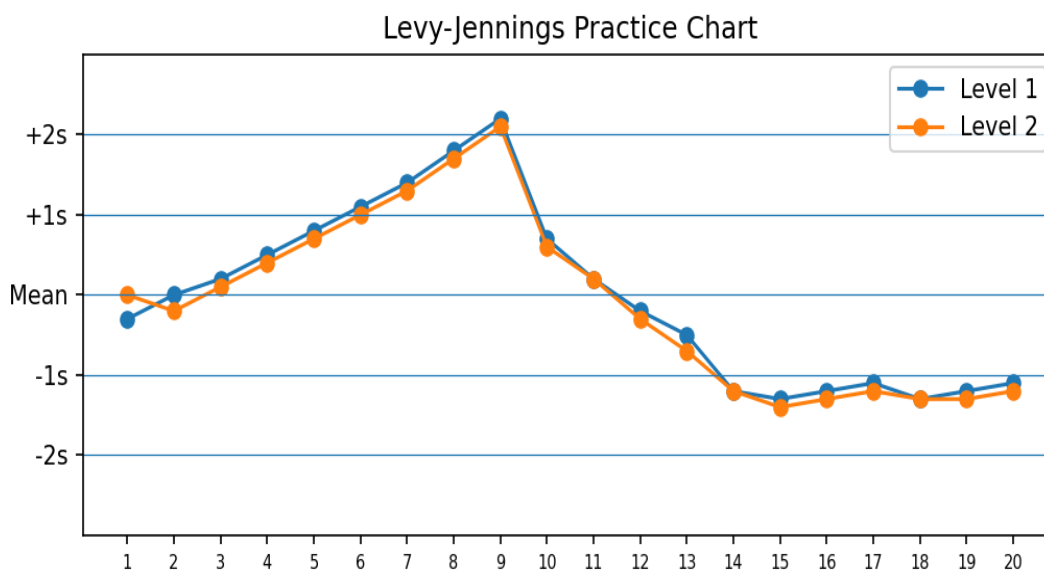
Part III - Graph and Graphic Interpretation

41. Using this normal-distribution graphic, about how many QC results lie outside ± 2 SD by chance?



- A. 5%
- B. 32%
- C. 0.3%
- D. 50%

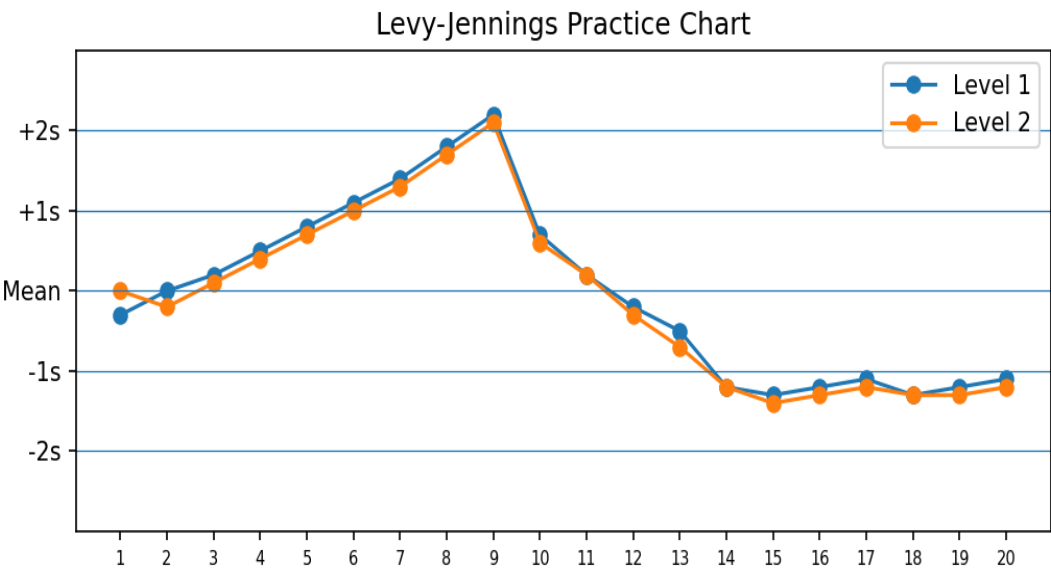
42. What pattern is evident around days 4-9?



- A. Trend
- B. Random scatter
- C. No change

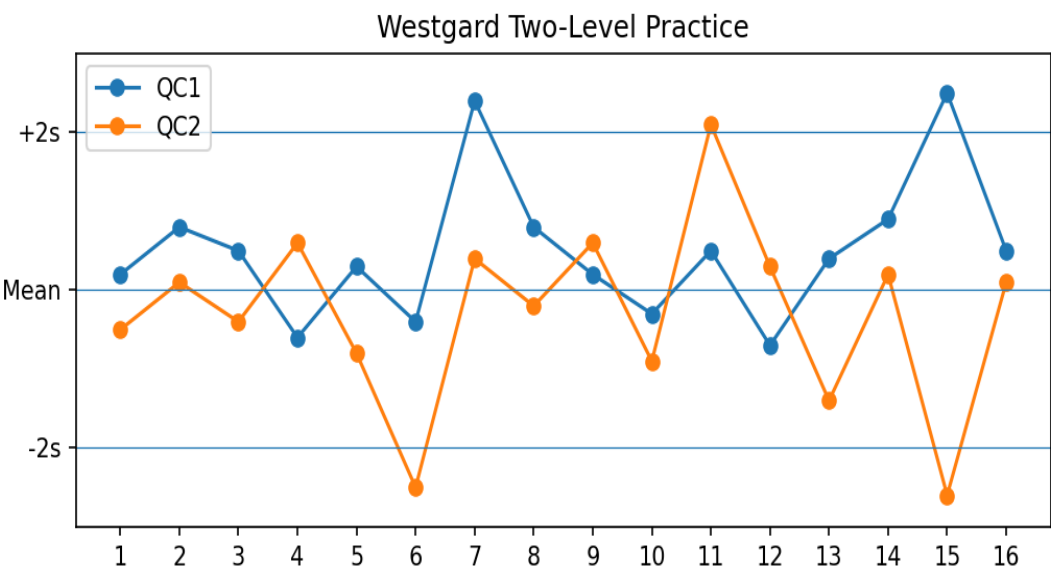
D. R4s only

43. What later pattern is evident when multiple points remain below the mean?



- A. Shift
- B. R4s only
- C. No problem
- D. Positive trend

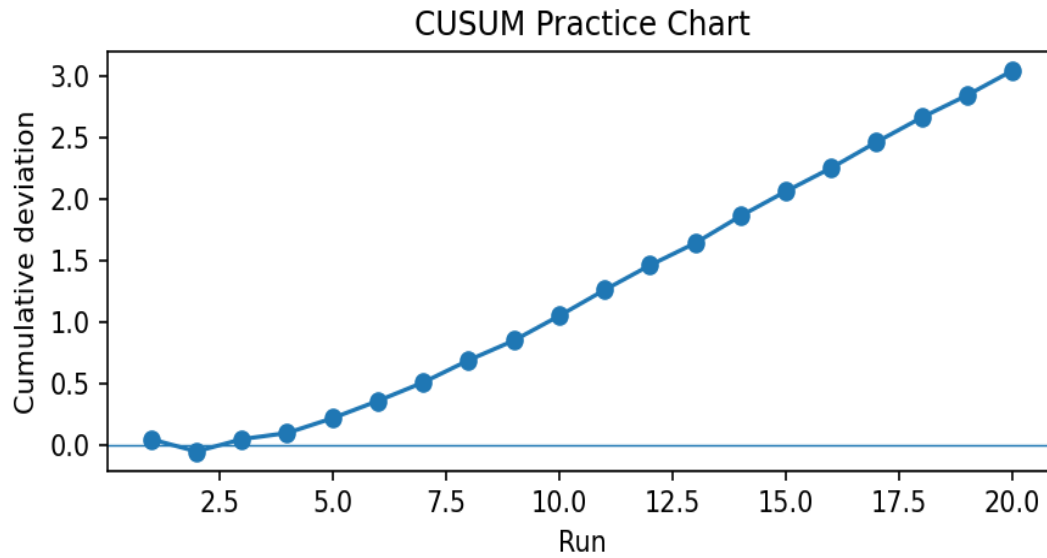
44. Which day most clearly violates R4s?



- A. 6
- B. 7
- C. 11

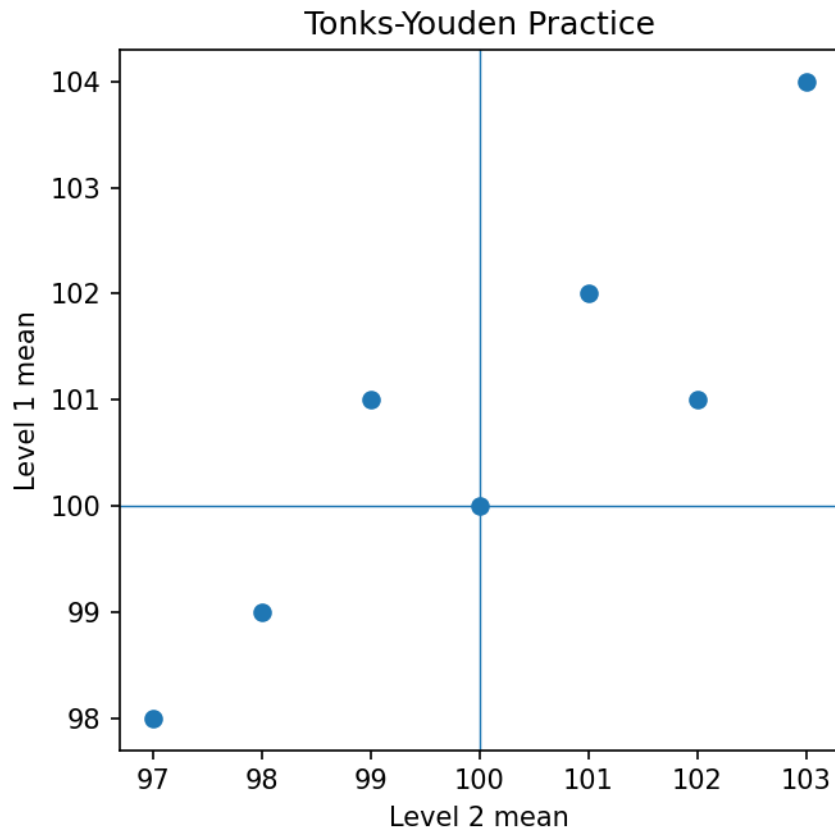
D. 15

45. What does the persistent positive CUSUM slope suggest?



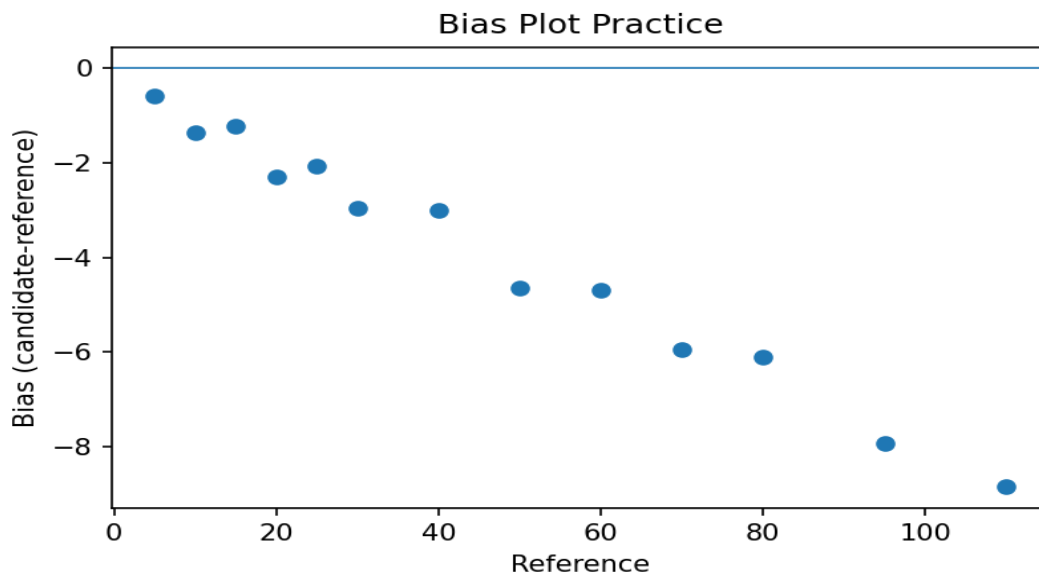
- A. Small sustained systematic movement
- B. Perfect random balance
- C. One isolated R4s event
- D. Reference interval error

46. A point in the upper-right quadrant of this Youden plot suggests:



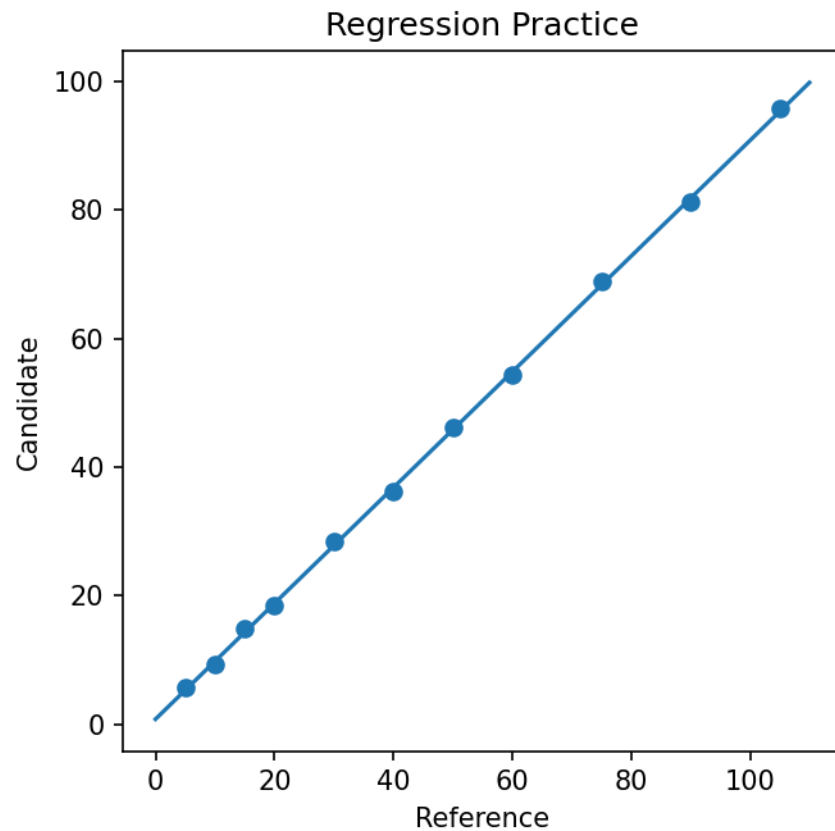
- A. Both control levels are positively biased
- B. Only random error
- C. One high and one low
- D. No bias

47. Interpret this bias plot.



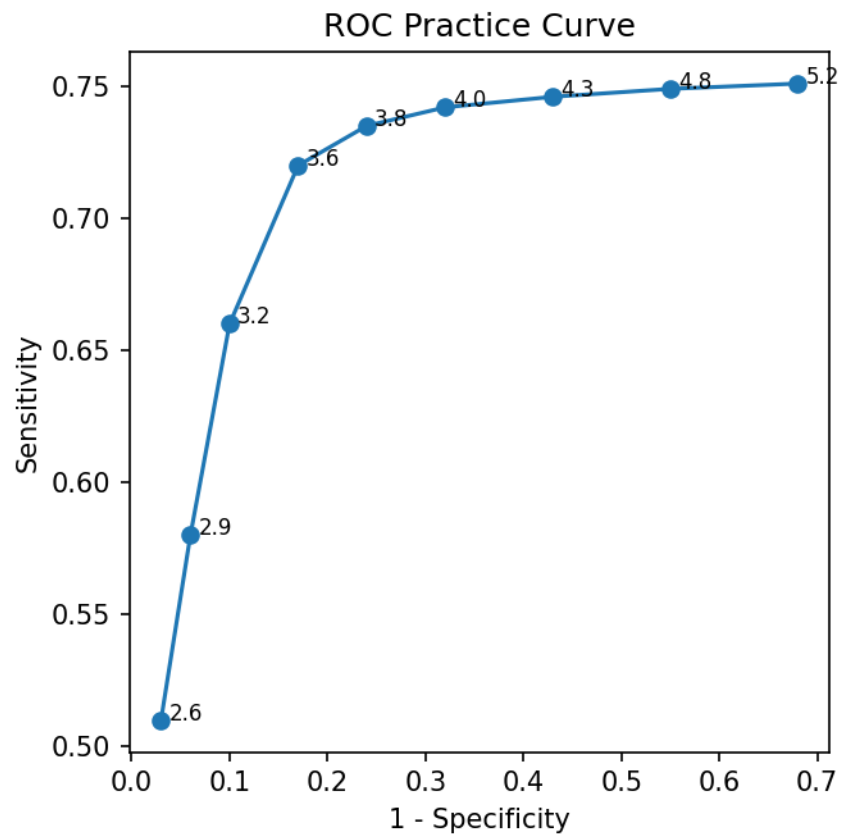
- A. Increasing negative proportional bias
- B. Constant positive bias
- C. No bias
- D. Random error only

48. What is the main disagreement in this regression plot?



- A. Proportional error
- B. Large constant error only
- C. No systematic error
- D. Specificity failure

49. Which labeled ROC cutoff is nearest the desirable upper-left region among the choices?



- A. 2.6
- B. 3.2
- C. 3.6
- D. 5.2

Answer Key & Rationales

1. C

Rationale: $4 \text{ g}/100 \text{ mL} \times 300 \text{ mL} = 12 \text{ g}$.

2. C

Rationale: $16 \text{ mL}/100 \text{ mL} \times 750 \text{ mL} = 120 \text{ mL}$.

3. A

Rationale: $C_1V_1=C_2V_2$ conserves the amount of solute during dilution.

4. B

Rationale: $\text{pH}=-\log[\text{H}^+]$, giving about 3.70.

5. C

Rationale: $\text{pH}+\text{pOH}=14$, so $\text{pH}=8.8$.

6. A

Rationale: Henderson-Hasselbalch relates pH, pKa and salt-to-acid ratio.

7. B

Rationale: $0.72/0.96 \times 240 = 180 \text{ mg/dL}$.

8. A

Rationale: The serum blank contribution is subtracted.

9. A

Rationale: This is the chapter's definition of an IU.

10. A

Rationale: CV normalizes SD to the mean.

11. C

Rationale: About 95.4% lie within ± 2 SD.

12. A

Rationale: 1_2s is triggered by one control outside 2 SD.

13. B

Rationale: 2_2s indicates systematic error.

14. A

Rationale: One result beyond 3 SD violates 1_3s .

15. A

Rationale: 4_1s detects systematic error.

16. A

Rationale: A trend is a steady directional change.

17. A

Rationale: A persistent displacement to one side is a shift.

18. A

Rationale: Levy-Jennings is used for routine QC monitoring.

19. A

Rationale: Tonks-Youden compares monthly means across labs.

20. A

Rationale: CUSUM is sensitive to small persistent changes.

21. A

Rationale: Student's t test assesses mean differences.

22. A

Rationale: F is the ratio of variances.

23. A

Rationale: Bias=candidate-reference=+6.

24. A

Rationale: Proportional error changes with concentration.

25. A

Rationale: The intercept estimates constant error.

26. A

Rationale: The slope estimates proportional error.

27. A

Rationale: Specificity is the fraction of disease-free people testing negative.

28. A

Rationale: Sensitivity is the fraction of diseased people testing positive.

29. D

Rationale: At least 120 healthy reference individuals are recommended.

30. A

Rationale: SDI standardizes the laboratory-to-peer difference.

31. A

Rationale: Delta checks compare current and prior patient values.

32. A

Rationale: Six Sigma quantifies defect frequency.

33. A

Rationale: A reagent-lot change may require validation with controls and patient samples.

34. A

Rationale: IQCP considers risk of error and severity of patient harm.

35. C

Rationale: $\text{Sigma} = (12 - 2) / 2 = 5$.

36. A

Rationale: CVs are 12%, 5%, 2%, and 2%; A is least precise.

37. B

Rationale: Candidate-reference = +11.

38. B

Rationale: $42 / (42 + 6) = 87.5\%$.

39. B

Rationale: $44 / (44 + 8) = 84.6\%$.

40. B

Rationale: A CV is about 1.11%; B is about 0.95%, so B has better relative precision.

41. A

Rationale: About 95.4% lie inside ± 2 SD, leaving roughly 5% outside.

42. A

Rationale: Both levels rise progressively, showing a trend.

43. A

Rationale: Sustained values on one side of the mean indicate a shift.

44. D

Rationale: Day 15 has one level above +2 SD and the other below -2 SD.

45. A

Rationale: Accumulated positive deviations indicate a small persistent shift/trend.

46. A

Rationale: Both levels exceed their peer means, supporting positive systematic bias.

47. A

Rationale: Bias becomes more negative as concentration rises.

48. A

Rationale: A slope below 1 indicates proportional underestimation.

49. C

Rationale: 3.6 gives the best balance of high sensitivity and relatively few false positives among these options.